

Operators in Bash

Definition:

Operators in Bash are special symbols or keywords used to perform operations on data such as arithmetic calculations, comparisons, logic, string tests, and file checks.

Types of Operators in Bash

1. Arithmetic Operators

Used for mathematical calculations.

Examples:

+ (Addition) $\$(a + b)$
- (Subtraction) $\$(a - b)$
* (Multiplication) $\$(a * b)$
/ (Division) $\$(a / b)$
% (Modulus) $\$(a \% b)$
** (Exponent) $\$(a ** b)$

2. Relational (Comparison) Operators

Used to compare numeric values.

Examples:

-eq Equal to
-ne Not equal to
-gt Greater than
-lt Less than
-ge Greater than or equal to
-le Less than or equal to

3. Boolean (Logical) Operators

Used to combine or invert conditions.

Examples:

! NOT

-a AND

-o OR

Modern syntax:

```
[[ $a -lt 10 && $b -gt 5 ]]
```

4. String Operators

Used to compare strings.

Examples:

= Equal

!= Not equal

< Less than (alphabetically)

> Greater than (alphabetically)

-z String is empty

-n String is not empty

5. File Test Operators

Used to test file types and properties.

Examples:

-e file True if file exists

-f file True if regular file

-d file True if directory

-r file True if readable

-w file True if writable

-x file True if executable

-s file True if file not empty

6. Assignment Operators

Used to assign or update variable values.

Examples:

= Assign

- `+=` Add and assign
- `-=` Subtract and assign
- `*=` Multiply and assign
- `/=` Divide and assign

Example:

```
#!/bin/bash
a=8
b=3
if [[ $a -gt $b && $b -ne 0 ]]; then
    echo "a is greater than b and b is not zero"
    echo "Sum: $((a + b))"
fi
```

Summary Table

Category	Examples	Used For
Arithmetic	<code>+, -, *, /, %, **</code>	Math operations
Relational	<code>-eq, -lt, -gt</code>	Compare numbers
Logical	<code>&&, , !</code>	Combine conditions
String	<code>=, !=, -z, -n</code>	Compare text
File Test	<code>-f, -d, -e</code>	File/dir checking
Assignment	<code>=, +=, -=, *=, /=</code>	Assign/update values